UDP Client Experiment

1. UDP Client Overview

UDP (User Datagram Protocol) is a **connectionless, unreliable but efficient** transport layer protocol. Unlike TCP, UDP does not establish connections, does not guarantee delivery, and does not guarantee ordering, but it has **low overhead and low latency**, making it very suitable for scenarios with high real-time requirements where occasional packet loss is acceptable.

In IoT applications, UDP is widely used for **high-frequency sensor data reporting** (such as temperature and humidity sampled 100 times per second where losing a few frames doesn't matter), **LAN device discovery** (broadcast/multicast probing for online devices on the same subnet), **DNS resolution** (small query volume, timeout-retry is enough), and **real-time audio/video streaming** (latency takes priority over image quality). Mastering UDP's `sendto()` / `recvfrom()` programming model is the foundation for later advanced experiments such as custom protocol packaging, real-time data reporting, and LAN device discovery.

In this experiment the ESP32-S3 acts as a UDP client, communicating bidirectionally with the PC network debugging assistant — active handshake, non-blocking command reception, periodic heartbeat, and support for commands such as `ping` / `status` / `led on` / `led off` / `who`.

Typical application scenarios:

Application Type	Example
Real-time data streams	High-frequency sensor sampling reporting (losing a few frames doesn't matter)
LAN discovery	mDNS / SSDP device discovery protocols
Voice/video	VoLTE, real-time audio/video (low latency > reliability)
Broadcast/multicast	Sending one message to multiple devices

> UDP requires no connection; you can send directly with `sendto()`. The peer address is obtained in each `recvfrom()`.

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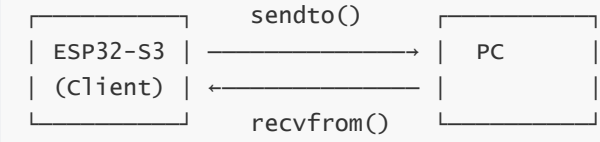
2. Core Concepts

2.1 UDP Communication Model



- UDP is connectionless; no `connect()` is needed; you send directly to the target address
- Each packet is independent; the sender and receiver do not need to maintain connection state
- `sendto(data, (ip, port))` sends, `recvfrom(bufsize)` receives and obtains the source address

2.2 Command Table

PC Command	ESP32 Reply	Description
`ping`	`pong`	Connectivity test
`status`	Runtime + free memory	Device status
`led on`	`LED ON (OK)`	Turn on GPIO48 LED
`led off`	`LED OFF (OK)`	Turn off GPIO48 LED
`who`	ESP32-S3 UID	Device unique ID
Other text	`echo: <original>`	Echo

2.3 Core API

```python
import socket

sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)  # UDP socket
sock.bind(("0.0.0.0", 8080))                # bind local port
sock.sendto(b"hello", ("192.168.11.183", 8080)) # send
data, addr = sock.recvfrom(256)              # recv + get sender
```

3. Hardware Dependencies

The ESP32-S3 has built-in WiFi. The PC network debugging assistant selects UDP mode, listens on port `8080`, and the target port is `8080` (the ESP32 bind port).

4. Project Architecture and Flow

```
Script execution (single-file .py)
  |
  |- wifi_connect(): connect to WiFi (STA mode)
  |- udp_socket(): create UDP socket -> bind(8080)
  |- udp_send(): handshake -> HELLO
  |
  `- while True:
       |- WiFi disconnect watchdog -> auto reconnect
       |- udp_recv(): non-blocking receive (0.3s timeout)
       |- handle_command(): command dispatch
       `- send heartbeat every 30s
```

5. Source Code Explained

Full source code: `Source Code -> MicroPython -> 4.Network Protocols -> UDP_Client -> UDP_Client.py`

5.1 Creating a UDP Socket and Handshake

`SOCK_DGRAM` = UDP; `bind` specifies the local port and then sends a HELLO handshake packet to the PC.

```
ESP32_PORT = 8080                # local bind port
PC_IP      = "192.168.11.183"     # PC IP address
PC_PORT    = 8080                # PC listen port

sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.bind(("0.0.0.0", ESP32_PORT))
sock.settimeout(0.3)             # non-blocking polling

udp_send(sock, PC_IP, PC_PORT, f"HELLO from ESP32-S3 [{esp_ip}]")
```

5.2 Non-Blocking Reception

`settimeout(0.3)` implements non-blocking behavior — if no data arrives within 0.3s, an OSError is raised; catching it returns None and continues the main loop.

```
def udp_recv(sock, bufsize=256):
    try:
        data, addr = sock.recvfrom(bufsize)
        return data, addr
    except OSError:
        return None, None          # timeout, no data
```

5.3 Command Dispatch

Supports five commands `ping` / `status` / `led on` / `led off` / `who`; unmatched commands echo the original content.

```
def handle_command(sock, addr, cmd):
    if cmd == "ping":
        udp_send(sock, *addr, "pong")
    elif cmd == "status":
        import gc
        reply = f"ESP32 up {time.time():.0f}s | free heap: {gc.mem_free()} B"
        udp_send(sock, *addr, reply)
    elif cmd == "led on":
        from machine import Pin; Pin(48, Pin.OUT, value=1)
        udp_send(sock, *addr, "LED ON (OK)")
    elif cmd == "who":
        import machine, ubinascii
        uid = ubinascii.hexlify(machine.unique_id()).decode()
        udp_send(sock, *addr, f"ESP32-S3 UID: {uid}")
    else:
        udp_send(sock, *addr, f"echo: {cmd}")
```

Command set:

Command	Reply	Description
`ping`	`pong`	Connectivity test
`status`	`ESP32 up Xs / free heap: X B`	Running status
`led on`	`LED ON (OK)`	Turn on GPIO48
`led off`	`LED OFF (OK)`	Turn off GPIO48
`who`	`ESP32-S3 / UID: xxx`	Device unique ID
Other text	`echo: <original>`	Echo if unmatched

5.4 Main Entry

`wifi_connect` connects to WiFi -> `udp_socket` creates and binds -> HELLO handshake -> main loop: receive commands + reconnect on disconnect + 30s heartbeat.

```
def main():
    wlan = wifi_connect(WIFI_SSID, WIFI_PASSWORD)  # connect to WiFi
    esp_ip = wlan.ifconfig()[0]

    sock = udp_socket(ESP32_PORT)                  # create socket
    udp_send(sock, PC_IP, PC_PORT, f"HELLO from ESP32-S3 [{esp_ip}]")

    while True:
        if not wlan.isconnected():                 # WiFi watchdog
            wlan = wifi_connect(WIFI_SSID, WIFI_PASSWORD)
        data, addr = udp_recv(sock)                # non-blocking recv
        if data:
            handle_command(sock, addr, fmt(data))  # dispatch
```

```
        if time.time() - last_hb >= 30:              # heartbeat every 30s
            udp_send(sock, PC_IP, PC_PORT, f"heartbeat [{esp_ip}]")
```

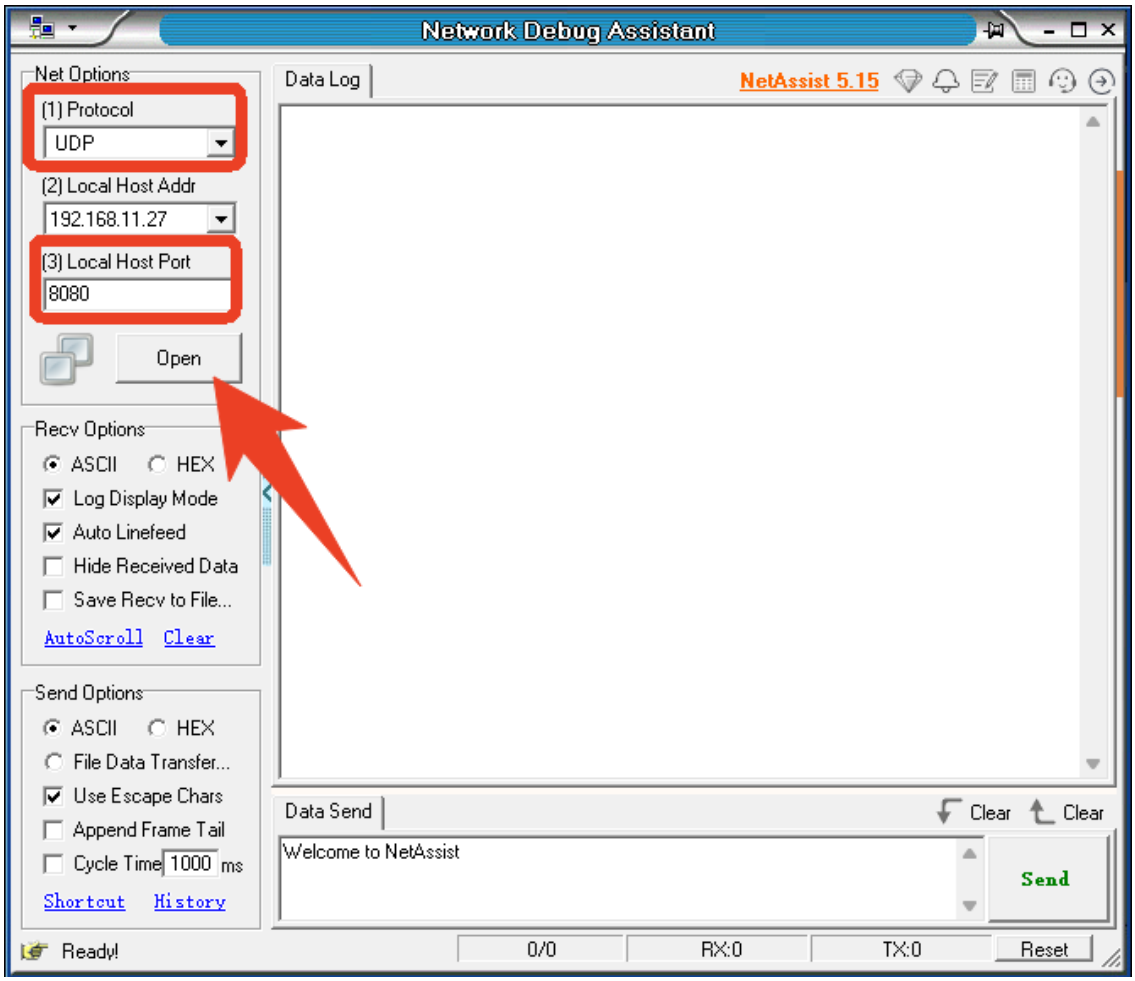
6. Operating Procedures

6.1 Environment Preparation

Thonny IDE steps: See `MicroPython -> 1. Before Development -> 3.Thonny IDE`.

1. PC network debugging assistant 

UDP mode, listen on port `8080`



2. Modify `WIFI_SSID` / `WIFI_PASS` / `PC_IP`

```
# ==================== 用户配置 / User Config ====================
WIFI_SSID     = "Yahboom2"          # WiFi名称 / WiFi SSID
WIFI_PASSWORD = "yahboom890729"     # WiFi密码 / WiFi password

PC_IP         = "192.168.11.27"   # PC-IP地址 / PC IP (ipconfig)
PC_PORT       = 8080              # PC网络助手监听端口 / PC listen port
ESP32_PORT    = 8080              # ESP32-本地绑定端口 / ESP32 local bind port
# ============================================================
```

6.2 Flashing and Running

Run `UDP_Client.py`; the ESP32 sends HELLO to the PC. The PC sends commands and the ESP32 replies.

6.3 Serial Output Example

```
==================================================
  ESP32-S3 UDP Client  <-->  PC Network Assistant
==================================================
[WiFi] Already connected. IP: 192.168.11.229
[SOCK] UDP bound to 0.0.0.0:8080
[TX] -> 192.168.11.27:8080  |  HELLO from ESP32-S3 [192.168.11.229]
--------------------------------------------------
  PC   listen : 192.168.11.27:8080
  ESP32 bind  : 192.168.11.229:8080
--------------------------------------------------
[RX] <- 192.168.11.27:8080  |  led off
[TX] -> 192.168.11.27:8080  |  LED OFF (OK)  # GPIO44=1
[RX] <- 192.168.11.27:8080  |  led on
[TX] -> 192.168.11.27:8080  |  LED ON  (OK)  # GPIO44=0
[TX] -> 192.168.11.27:8080  |  heartbeat [192.168.11.229]
[RX] <- 192.168.11.27:8080  |  who
[TX] -> 192.168.11.27:8080  |  ESP32-S3  |  UID: a4cb8fc6b864
[RX] <- 192.168.11.27:8080  |  ping
[TX] -> 192.168.11.27:8080  |  pong
[RX] <- 192.168.11.27:8080  |  status
[TX] -> 192.168.11.27:8080  |  ESP32 up 838500500s | free heap: 120352 B
```

Network debugging assistant print:

```
[2026-07-27 20:46:52.170]# SEND ASCII/7 to 192.168.11.229 :8080 >>>
led off

[2026-07-27 20:46:52.280]# RECV ASCII/24 from 192.168.11.229 :8080 <<<
LED OFF (OK)  # GPIO44=1

[2026-07-27 20:46:58.681]# SEND ASCII/6 to 192.168.11.229 :8080 >>>
led on

[2026-07-27 20:46:58.783]# RECV ASCII/24 from 192.168.11.229 :8080 <<<
LED ON  (OK)  # GPIO44=0

[2026-07-27 20:47:17.727]# RECV ASCII/26 from 192.168.11.229 :8080 <<<
heartbeat [192.168.11.229]

[2026-07-27 20:47:17.889]# SEND ASCII/3 to 192.168.11.229 :8080 >>>
who

[2026-07-27 20:47:17.977]# RECV ASCII/30 from 192.168.11.229 :8080 <<<
ESP32-S3  |  UID: a4cb8fc6b864

[2026-07-27 20:47:33.034]# SEND ASCII/4 to 192.168.11.229 :8080 >>>
ping

[2026-07-27 20:47:33.131]# RECV ASCII/4 from 192.168.11.229 :8080 <<<
pong

[2026-07-27 20:47:42.020]# SEND ASCII/6 to 192.168.11.229 :8080 >>>
status

[2026-07-27 20:47:42.249]# RECV ASCII/41 from 192.168.11.229 :8080 <<<
ESP32 up 838500500s | free heap: 120352 B
```

6.4 Verification Methods

Operation	Expected Result
PC sends `ping`	Replies `pong`
PC sends `led on`	GPIO48 LED turns on, replies `LED ON (OK)`
PC sends `who`	Replies the ESP32-S3 unique ID

7. Common Problems

Symptom	Cause	Solution
PC cannot receive HELLO	Target IP/port mismatch	Confirm `PC_IP` and `PC_PORT` match the PC
ESP32 cannot receive commands	Wrong target port when the PC sends	The target port should be `8080` (the ESP32 bind port)
Occasional packet loss	UDP does not guarantee delivery	Normal; handle resending at the application layer